Structure and Function of the Respiratory System

NURA 340 Pathophysiology | Lecture Study Guide

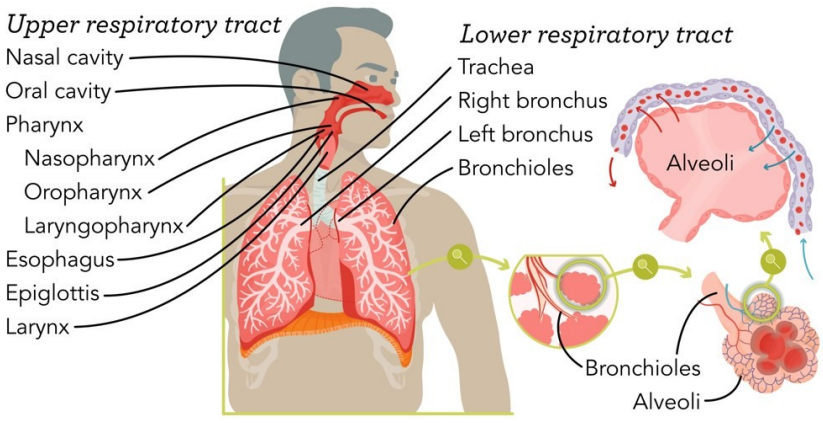
Instructor: Sarah Mortensen, MSN, RN, CNE, CCRN

Lecture roadmap: Upper vs. lower tract → Respiratory muscles → Pleural membranes → Ventilation vs. oxygenation → Physiology of ventilation → Gas exchange and transport → Regulatory mechanisms → Protective mechanisms → Environmental factors

The respiratory tract does not operate in isolation. It is tied to three core pathophysiological domains: acid-base imbalance, fluid dynamics, and immune defense and inflammation. Keep these connections in mind as you move through both the anatomy and the mechanics of breathing.

Upper vs. Lower Respiratory Tract

	Upper tract	Lower tract
Structures	Nasal (or oral) cavity, paranasal sinuses, pharynx (nasopharynx, oropharynx, laryngopharynx), epiglottis, larynx	Trachea, right and left primary (mainstem) bronchi, bronchioles, terminal alveoli
Function	Primary gateway and filter for airborne pathogens; warms, humidifies, and filters incoming air. Immune defense	Conducting airways transition to the delicate alveolar-capillary unit. Gas exchange and advanced local defense
Disease effects	Viral rhinitis or pharyngitis cause localized inflammation (erythema, edema, mucus hypersecretion) that keeps pathogens from descending into the sterile lower tract	Pneumonia or ARDS disrupt fluid balance and impair gas exchange, leading to systemic acid-base imbalance



Upper and lower respiratory tract (lecture slide)

Upper Tract Immune Function

The nasal mucosa is rich in vascularized mucosal glands. When irritants or infectious agents enter, local mast cells and macrophages release histamine and inflammatory cytokines. This increases mucosal blood flow and capillary permeability, producing nasal congestion, rhinorrhea (runny nose), and edema. The localized vascular expansion acts as a physical and immunologic barrier that neutralizes pathogens before they migrate deeper into the tracheobronchial tree.

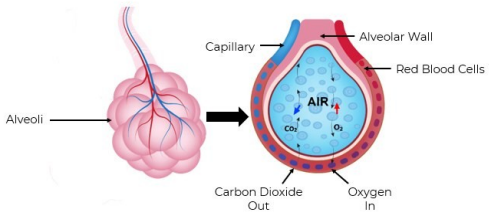
For anything that gets past the nasal mucosa, a key defense is the mucociliary escalator (see Protective Mechanisms).

Lower Tract Structure and Fluid Balance

The trachea branches into the mainstem bronchi, which divide into smaller and smaller bronchioles and end in hundreds of millions of thin-walled alveoli. The alveoli are the functional unit where O_2 diffuses into the pulmonary capillaries and CO_2 diffuses out.

> **Acid-base link:** Any change in alveolar surface area or membrane thickness directly changes systemic pH by altering arterial CO_2 pressure ($PaCO_2$).

The lower airway relies on tight junctions and intact alveolar membranes to keep the alveoli dry so gas exchange can occur. When bacteria or viruses reach the alveoli, local inflammation increases capillary permeability and floods the alveolar space with exudate fluid, impairing diffusion. Macrophages neutralize pathogens and particles that bypass upper tract defenses, which helps, but ideally the lungs stay clear.



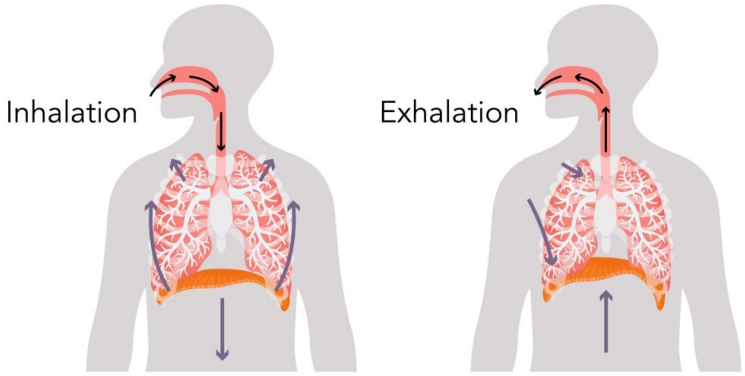
Alveolus and capillary (lecture slide)

Respiratory Muscles

Breathing relies on dynamic pressure changes created by the lungs, diaphragm, intercostal muscles, and rib cage, which work together as a mechanical pump.

- **Intercostal muscles:** between the ribs; assist with chest wall movement.
- **Accessory muscles:** in the neck and upper chest; support breathing during increased demand such as illness or exertion. This is what you see as retractions, when muscles work harder and the chest wall sucks in with each breath.

	Inhalation	Exhalation
Muscles	Diaphragm contracts and moves downward; external intercostals lift the rib cage upward and outward	Passive relaxation of the diaphragm and intercostals (elastic recoil of the lungs)
Thoracic volume	Increases	Decreases
Pressure (Boyle's law)	Volume up, so intrapulmonary pressure drops below atmospheric. The negative gradient draws air in	Intrapulmonary pressure rises above atmospheric and pushes air out



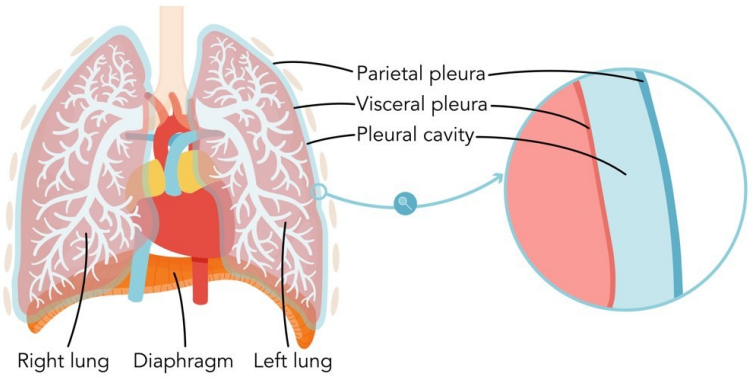
Inhalation vs. exhalation (lecture slide)

Acid-base link: Ventilation rate and depth determine CO_2 clearance. If muscle fatigue causes hypoventilation, CO_2 is retained, which drives the carbonic acid equation to the right and causes respiratory acidosis.

Pleural Membranes

The pleural membranes surround each lung and are crucial to pulmonary mechanics, fluid balance, and host defense.

- **Visceral pleura:** directly coats the lung parenchyma.
- **Parietal pleura:** lines the inner chest wall.
- **Pleural cavity:** the thin fluid-filled space between them. About 10–20 mL of serous fluid lubricates the surfaces so the lungs glide against the chest wall and friction is reduced. With too little lubricant the lungs do not move smoothly and abnormal sounds may be heard.
- **Fluid balance:** maintained by systemic capillary hydrostatic pressure, plasma oncotic pressure, and lymphatic drainage, so there is neither too much nor too little fluid.
- **Negative pressure:** the sealed pleural space keeps the elastic lungs adhering to the chest wall and prevents atelectasis (lung collapse). If this seal is disrupted, excess air or fluid enters the space.



Pleural membranes (lecture slide)

Condition	What it is
Pleural effusion	Excess pleural fluid from inflammation increasing capillary permeability. Usually collects at the bases of the lungs because of gravity
Empyema	Infected fluid accumulating in the pleural space

Ventilation vs. Oxygenation

Strictly differentiate these two. When you reach nursing interventions, ask whether an intervention helps oxygenation or ventilation. Most respiratory disorders are ventilation problems, so we mainly support patients through ventilation; we can help oxygenation a little, but that alone does not solve the underlying problem.

	Ventilation	Oxygenation
Definition	Mechanical, neuromuscular movement of air in and out of the lungs	Passive movement of oxygen from the alveoli across the alveolar-capillary membrane into the blood, where it binds hemoglobin
Affects	Blood gases; removal of CO_2	Oxygen delivery to the blood
Failure	Hyperventilation blows off excess CO_2, raising pH: respiratory alkalosis. Hypoventilation retains CO_2, lowering pH: respiratory acidosis	Alveolar-capillary membrane damaged or thickened by inflammatory exudate, pulmonary edema fluid, or fibrotic changes

Physiology of Ventilation

Pressures

- **Intrapleural pressure:** about −10 cm H_2O during quiet breathing. During inspiration the diaphragm contracts and expands the thoracic cavity, lowering the pressure and drawing air into the lungs.

- **Increased demand:** abdominal and intercostal muscles contract to push air out more forcefully, raising thoracic pressure as high as 80 cm H_2O.

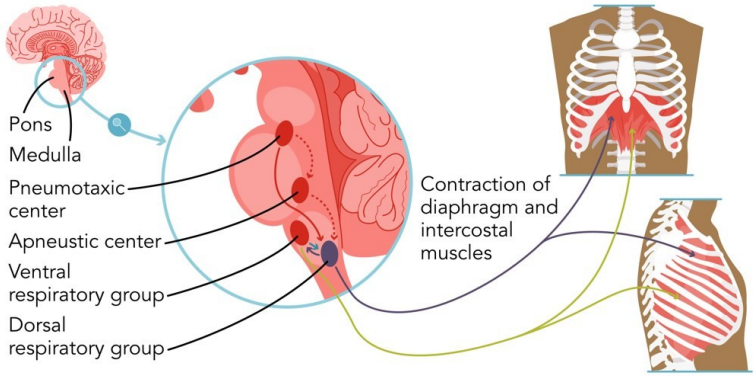
Lung Volumes

Volume	Definition
Tidal volume (TV)	Air inspired or expired in a normal quiet breath; about 500 mL in an adult
Inspiratory reserve volume (IRV)	Maximum air that can be forcefully inhaled beyond a normal tidal inhalation (a really deep breath in)
Expiratory reserve volume (ERV)	Maximum air that can be forcefully exhaled after a normal tidal exhalation
Residual volume (RV)	Air remaining in the lungs after maximal forced exhalation. It keeps the alveoli from collapsing completely with each breath; otherwise they would have to be popped open every time, taking more pressure and effort

Obstructive disease: In asthma or COPD, airway edema and bronchial spasm trap air and increase residual volume. This reduces effective alveolar ventilation, leading to chronic CO_2 retention and altered acid-base homeostasis.

Role of Respiratory Muscles

At rest, the diaphragm and external intercostals are enough, and exhalation is passive from elastic recoil. When ventilation demand rises (typically with illness, sometimes exertion), the primary muscles cannot manage alone and the accessory muscles help.



Respiratory muscles and neural control (lecture slide)

Compliance and Resistance

	Meaning	Clinical picture
Compliance	Ease with which the lungs and chest wall expand	Low compliance means stiff lungs (acute inflammatory conditions, pulmonary edema), which increases the work of breathing
Resistance	Friction encountered by airflow	Airway inflammation, mucus accumulation, and bronchial constriction dramatically raise resistance

Both are hard to measure, but you can feel them when manually bagging a patient with an Ambu bag: if you must press harder to get air in, compliance is low or resistance is high.

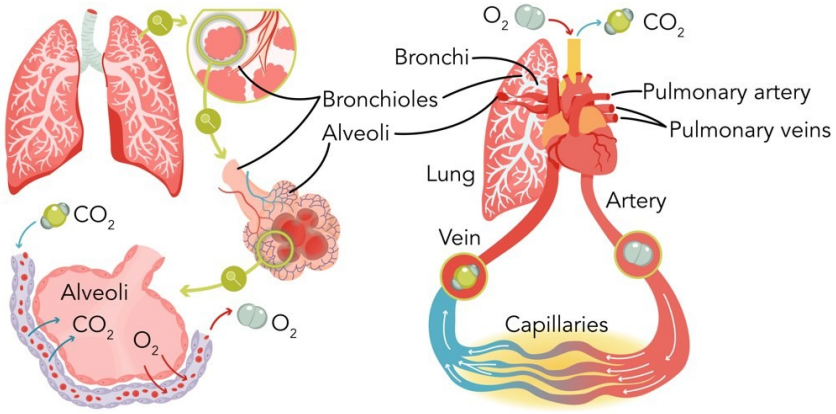
Neural Control

Respirations are regulated automatically in the brain stem. Neural centers respond to input from chemoreceptors and mechanoreceptors, generate rhythmic breathing, and control the diaphragm and intercostals, so we do not have to think about breathing. Brain injury can disrupt this control, and prolonged acid-base imbalance can change how well the brain senses pH changes.

Gas Exchange and Transport

Diffusion

Once ventilation brings air to the alveoli, gas exchange occurs by passive diffusion across the ultrathin alveolar-capillary membrane. Oxygen moves down its gradient from the high partial pressure in the alveoli to the capillary blood, while carbon dioxide moves in the opposite direction.

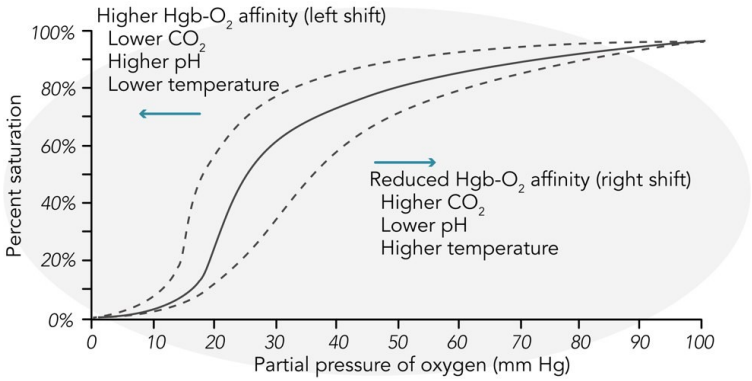


Gas exchange at the alveoli (lecture slide)

Oxygen Transport

- Over 98% of blood oxygen is carried bound to hemoglobin; less than 2% is dissolved in plasma. This is why adequate blood volume matters.
- Hemoglobin's affinity for oxygen shifts with the metabolic and acid-base environment of the tissues.

- **Easier release (right shift):** fever, infection, and hypermetabolism (higher CO_2, lower pH, higher temperature) drive oxygen off hemoglobin so it can enter ischemic tissues.
- **Harder release (left shift):** low CO_2 (higher pH) and lower temperature.



Oxygen-hemoglobin dissociation curve shifts (lecture slide)

Carbon Dioxide Transport

Form	Share	How
Bicarbonate ions	~70%	CO_2 is converted inside red blood cells by carbonic anhydrase
Carbaminohemoglobin	~20%	Binds directly to the amino groups of hemoglobin
Dissolved in plasma	~10%	Travels as dissolved gas, which exerts the partial pressure

Partial Pressures

Arterial partial pressures of O_2 and CO_2 determine the direction of gas movement, from high pressure to low pressure.

Measure	Expected range
PaO_2	75–100 mmHg
$PaCO_2$	35–45 mmHg (same as on blood gases)

Know this: If PaO_2 falls below about 60 mmHg, hemoglobin saturation drops steeply, limiting oxygen delivery to tissue because there is not enough oxygen to saturate the hemoglobin.

Factors Affecting Gas Exchange: V/Q

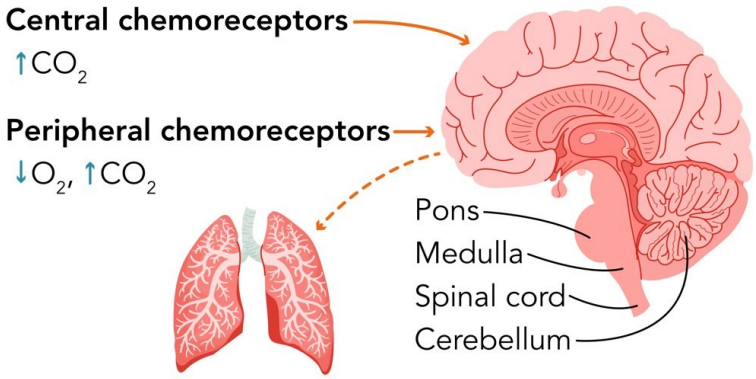
The ventilation-perfusion (V/Q) ratio determines the efficiency of gas exchange. The question is whether there is enough blood flow and enough oxygen reaching the tissues. Any V/Q mismatch impairs gas exchange.

	Dead space	Shunt
Problem	Inadequate perfusion	Inadequate ventilation (insufficient oxygen reaching alveoli) despite adequate perfusion
Example	Pulmonary embolism: a clot occludes blood flow to an area of lung even though oxygen is plentiful	Alveoli filled or collapsed so blood passes by with no gas exchange

Regulatory Mechanisms

The brain controls breathing through tight chemical feedback loops.

Receptor	Location	Senses
Central chemoreceptors	Medulla	Hydrogen ion concentration in cerebrospinal fluid, which reflects arterial CO_2. Elevated CO_2 is the primary driver of normal respiratory drive
Peripheral chemoreceptors	Carotid bodies and aortic arch	Arterial blood: decreased PaO_2, increased CO_2, and increased hydrogen ions



Central and peripheral chemoreceptors (lecture slide)

Hypoxic Drive

In chronic CO_2 retention, such as severe end-stage COPD, central chemoreceptors become desensitized over time. The body shifts its main breathing stimulus from CO_2 to the peripheral chemoreceptors responding to low arterial oxygen. This is the hypoxic drive.

Updated thinking: This was once emphasized heavily: the fear was that giving more oxygen than needed to these patients would remove their stimulus and make them stop breathing. With more experience caring for COPD patients, we have learned this can happen but is not as dramatic as once thought.

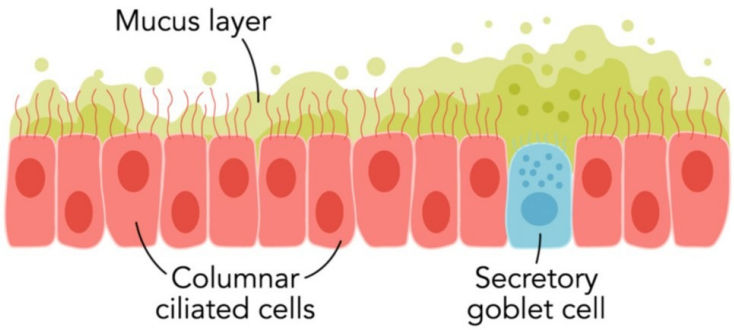
Protective Mechanisms

The lungs are continuously exposed to ambient air, so robust physical protective mechanisms are needed to prevent lower tract infection and inflammation.

Mucociliary Escalator

Mucus traps inhaled particles, and cilia move it up the airways toward the pharynx, where it is swallowed (into the stomach) or expelled, most often by coughing or sneezing.

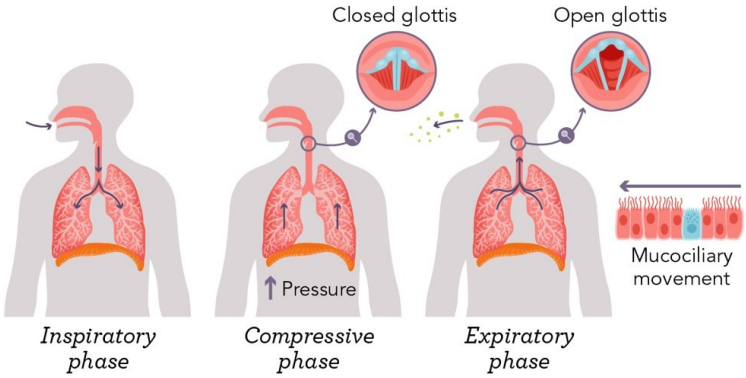
- **Dehydration thickens mucus.** If a patient has thick mucus, keep them well hydrated to help thin it.
- **Cigarette smoke and viral infections can paralyze ciliary motion,** which is why these patients are at high risk of bacterial infection.



Mucus layer, ciliated cells, and goblet cells (lecture slide)

Cough Reflex

Triggered when mechanical or chemical irritants activate vagal sensory receptors in the larynx, trachea, or carina. It clears trapped pathogens and secretions from the airways, which is beneficial for anyone who is sick or has excess mucus.

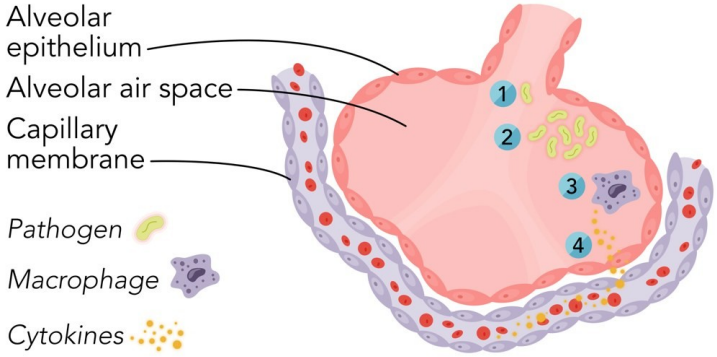


Phases of a cough (lecture slide)

Immune Defenses and Air Filtration

- **Nasal hairs:** filter large particles.
- **Turbinates:** the bony structures create turbulent airflow, driving particles against the sticky mucous membranes.

- **Tonsils and adenoids:** peripheral lymphoid structures that present inhaled antigens to lymphocytes. This is one of the first places lymphocytes meet antigens from pathogens entering the body.
- **Alveolar macrophages:** neutralize pathogens and particles that reach the alveoli.



Alveolar immune response (lecture slide)

Environmental and Lifestyle Factors

Factor	Effect
Irritants (tobacco, pollution, allergens)	Chronic exposure triggers persistent low-grade airway inflammation. It alters goblet cell function, destroys alveoli, and impairs the mucociliary escalator, predisposing to recurrent lower respiratory infections
Physical activity	Raises metabolic demand, increasing CO_2 production and O_2 consumption. The respiratory center compensates by increasing tidal volume and rate to preserve acid-base balance. Healthy activity strengthens the respiratory system, but when sick, extra demand can fatigue the body so much it cannot fight the illness
Altitude	Lower atmospheric pressure lowers the partial pressure of inspired oxygen. Acute hypoxia stimulates peripheral chemoreceptors, causing hyperventilation, which blows off CO_2 and causes respiratory alkalosis until renal compensation restores bicarbonate balance. Travelers to high altitude (altitude sickness), and people moving from high to low altitude, may not be adapted

Quick Review

- Ventilation = air movement and CO_2 clearance. Oxygenation = O_2 crossing the membrane onto hemoglobin. Most disorders are ventilation problems.
- Hypoventilation: respiratory acidosis. Hyperventilation: respiratory alkalosis.
- Inhalation is active (diaphragm and external intercostals); exhalation is passive.
- Tidal volume ~500 mL; residual volume keeps alveoli open and rises with obstructive disease.
- 98% of O_2 is on hemoglobin; CO_2 is ~70% bicarbonate, ~20% carbaminohemoglobin, ~10% dissolved.

- PaO_2 75–100 mmHg; $PaCO_2$ 35–45 mmHg; saturation falls steeply below ~60 mmHg PaO_2.
- Dead space = no perfusion (PE); shunt = no ventilation.
- CO_2 (via central chemoreceptors in the medulla) is the main drive; low O_2 (peripheral chemoreceptors) drives breathing in chronic CO_2 retainers.
- Dehydration thickens mucus; smoke and viruses paralyze cilia.

Notes on Transcript Cleanup

Corrected mis-transcriptions: "mucosal/mucilary escalator" is the mucociliary escalator; "bony turbulence created the turbulent blood flow" is read as turbinates creating turbulent airflow; "corona" is read as carina; "drawing air out of the lungs" during inspiration is read as drawing air in; "higher CO_2, sorry" was a self-correction and is kept as CO_2 retention. The lecture gives intrapleural pressure as about −10 cm H_2O for both quiet inspiration and expiration; it actually varies between the two, so check your notes. The right-shift/left-shift explanation was assembled from the lecture and the slide curve. The lecture defines a shunt only loosely, so I added a standard example (alveoli filled or collapsed, so blood passes by without gas exchange); verify against your notes.